

EXECUTION OF PERI-OPERATIVE THEATRE SAFETY PRECAUTIONS

Unit of learning code- HE/OS/TT/CR/01/5/A

Related Unit of Competency in Occupational Standard;

Introduction to the unit of learning

This unit describes the competencies required to execute Perioperative Theatre safety precautions. It involves wearing of personal protective Theatre protective gears, verifying theatre checklist, patient care, proper handling of surgical instruments and consumables as well as responding to alarms in an Operation Theatre, followed in the context of theatre setting.

Summary of Learning Outcomes

1. Plan and prepare for Operation Theatre safety
2. Verify perioperative Theatre checklist
3. Care for patient and theatre team
4. Handle instruments, apparatus and consumables
5. Respond to perioperative Theatre alarms
6. Evaluate, wind-up Operation Theatre safety

Learning Outcome 1: Plan and prepare for Operation Theatre safety

Introduction to the learning outcome

This learning outcome specifies the content of competencies required to plan for Execution of Perioperative Theatre Safety Precautions. It includes Operation Theatre protective gears, Use of perioperative Theatre protective gears, perioperative Theatre safety precautions, Occupation health and safety in perioperative Theatre and Perioperative Theatre Practice.

Performance Standard

1. Protective gears are identified according to WHO standards
2. Protective gears are used according to WHO standards

Information Sheet

Definition of terms

Protective gears- Types of equipment that is worn to increase an individual's safety while performing potentially hazardous tasks

Theatre Safety precautions- Measures taken or observed in theatre to ensure that something with potential risk is safe

Laser (Light Amplification by the Stimulated Emission of Radiation) surgery- use of special light beams in surgery instead of instruments for making incisions on tissue

Volatile- something, such as a gas or liquid, that easily evaporates in normal temperature.

Electrosurgery (diathermy)- use of electricity in surgery to cut or coagulate tissues

Operation theatre protective gears

These are clothing or equipment that protects the wearer from direct contact with hazardous chemicals or potentially infectious body fluids. They may be worn for purposes of self protection or to prevent cross-infection in theatre. They include: disposable surgical caps, scrub trousers and tops, jackets, disposable shoe, surgical shoes; surgical masks, gowns, gloves, masks, aprons and eye protection. Institutional policies, national health guidelines and the World Health organization (WHO) recommendations determine the type of protective gear to be worn in the operation theatre. The type of gear to be worn may vary from one surgical procedure to the other depending on the risk of exposure to body fluids and respiratory droplets. All surgical practitioners in the operation theatre should take full responsibility over self and others in order to ensure surgical attire compliance.

Use of perioperative Theatre protective gears

Clothing and Covering for clothing

These serve as part of contact precautions based on Infection prevention and control (IPC) principles. Surgical scrubs should be worn as regular on-duty uniform in the operation theatre. Fluid-resistant aprons or gowns should be worn on top to reduce the likelihood of infected body fluids penetrating and contaminating the underlying clothes where splashing of blood or fluids is anticipated.

Footwear

Floors and surfaces may be contaminated with faeces and body fluids. Foot wear also provides optimal protection when floors are wet and rubber boots can protect against sharps injuries. Foot wear includes boots, closed shoes that fully cover the dorsum of the foot without laces, clogs. Shoe covers, preferably impermeable, should ideally be used over closed shoes to facilitate decontamination in procedures with body fluid splashes.

Head cover

It protects the skin and hair of the head and neck from contamination and the possibility of subsequent cross-infection. Hair and hair extensions need to fit inside the head cover. Single-use head covers are preferred.

Mucous membrane protection

Mucous membranes include the eyes, mouth and nose. Face shield, goggles and face masks should be used to completely cover them as an integral part of standard and droplet precautions (IPC measures that aim to prevent infection with pathogens that can be transmitted by large-particle droplets).

Face shields and goggles should not be used together, as this does not offer additional protection and causes more discomfort and fogging affecting vision.

Goggles: Good seal with the skin of the face to protect the eyes from fluid splashes and infectious droplets.

Face shield: Made of clear plastic and completely covers the sides and length of the face. It protects the eyes, nose and mouth's mucous membranes.

Surgical mask: It protects the nasal and oral mucosa from splashes and droplets of infectious material.

Gloves

They prevent self and cross infection; and reduce risk from needle-stick injuries. Sterile Surgical Gloves or High Level Disinfected Surgical Gloves should be worn during all procedures in which the main aim is to avoid introduction of pathogens into the patient. Single use examination gloves/ clean gloves should be worn for all procedures in which one will be in contact with intact mucous membranes and where the primary purpose of wearing gloves is to reduce the risk of being exposed to blood or other body fluids. The best IPC practice dictates that gloves should be changed between patients.

Video link on Donning PPES in theatre

https://www.youtube.com/watch?v=44eGL_dB4jk

Video link on Donning and Doffing of PPEs in surgical aerosol generating procedures

<https://www.youtube.com/watch?v=1M9lWuLzV28>

Theatre Safety protective gears

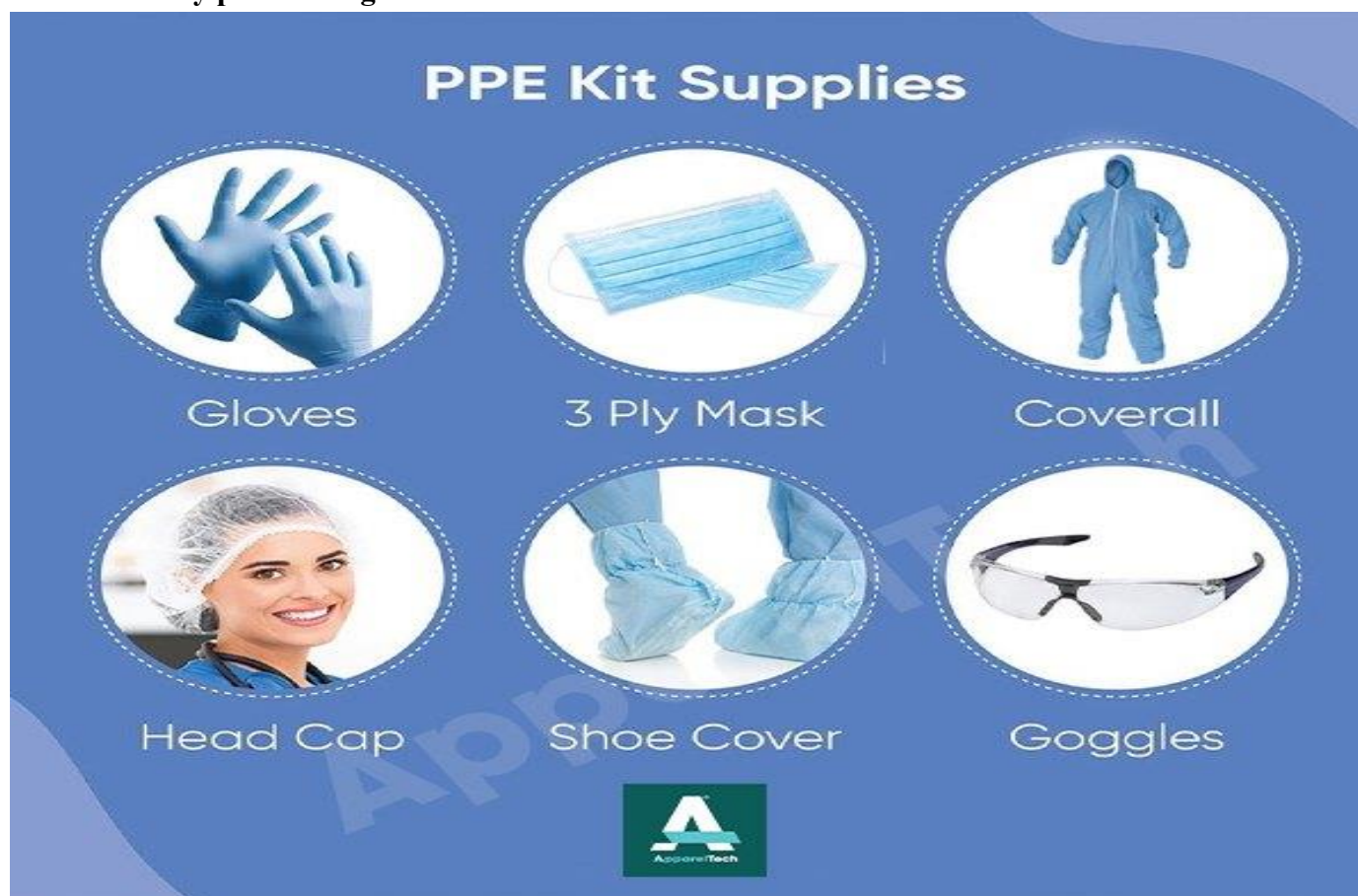


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Theatre Gear



Image

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Figure1-1.pn

Theatre protective clothing



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Perioperative theater safety precautions

Protecting patients from injury is one of the major roles of the perioperative theatre technicians.

The following are recommended practices for patient safety goals by the Joint Commission:-

- Improve the accuracy of patient identification
- Improve effectiveness of communication among caregivers
- Improve safety of using medications
- Reduce the risk of health care–associated infections
- Accurately and completely reconcile medications across continuum of care
- Reduce the risk of patient harm resulting from falls
- Reduce the risk of influenza and pneumococcal disease in institutionalized older adults
- Reduce the risk of surgical fires
- Implement applicable National Patient Safety Goals and associated requirements by components and practitioner sites
- Encourage patients' active involvement in their own care as a patient safety strategy
- Prevent health care–associated pressure ulcers (decubitus ulcers)
- Identify safety risks inherent in the organization's patient population
- Improve recognition and response to changes in a patient's condition

A perioperative theatre technologist has the responsibility of ensuring that the above goals are met, either directly or indirectly through collaboration with other health care members in order to maintain patient safety

Occupation health and safety in perioperative Theatre

During an operation health and safety procedures affect all those present including: the patient; the surgeon and the surgeon's assistants; the anaesthetist; nursing staff and theatre practitioners; health care assistants; any visitors to the theatre such as students and medical representatives. The following are health and safety issues specific to theatres:-

Special consideration must be given to patients under general anaesthetic, as they are unable to take care of their own safety needs or voice any concerns.

Pressure area care is a fundamental skill in the operation theatre because immobility which could be resulting from factors such as use of anesthetic agents or tissue incisions pose an additional risk factor to this.

Staff must ensure that visitors to theatres receive instruction in health and safety procedures. To ensure efficient and safe running of the theatre list, it should be managed by a designated person, usually the most senior. There should be sufficient staff on duty, and they should all have a clear understanding of their role in the team, based on their skills and abilities

All equipment should be tested before it is needed, and care should be taken with electrical equipment so that cables do not present a trip hazard or run across areas where there is a risk of fluids pooling. The heating should be set at a level that is safe and comfortable for patients and staff

A meticulous theatre-cleaning regime is fundamental to preventing infection. Theatres should be clean and dust free, ventilation systems must be in working order and doors must be kept closed.

The theatre floor should be cleaned thoroughly every day, and floors and surfaces should be damp-dusted between operations, ensuring any spillage of blood or body fluids is removed. A cleaning agent of proven activity should be used, cloths should be disposable and mop heads should be sterilised daily.

Although wearing gloves does not prevent accidental needlestick injuries, the DoH (1998b) suggests that they help reduce the volume of blood to which the wearer is exposed. Clean gloves should be put on by circulating staff before touching patients or any equipment, specimens, or drapes that could be contaminated.

Safety in use of specialist equipment

Electrosurgery (diathermy)

The patient's skin should be evaluated before and after diathermy use.

The patient should not be in contact with grounded metal objects as the provision of an alternative pathway for the electrical current could result in a burn.

Any metal parts of the operating table that could come into contact with the patient should therefore be padded and jewellery and piercings should be removed before arrival in theatre.

The return electrode should be placed over a vascular, muscular area to promote electrical conductivity and to dissipate heat.

Shaving may be necessary as hair at the dispersive site prevents complete plate contact with skin.

The plate should not be placed over bony prominences (this can impede return current), over implanted prostheses (there is a potential for overheating), over scar tissue or over areas distal to tourniquets where adequate tissue perfusion cannot be guaranteed.

Laser surgery

When lasers are in use theatre staff should wear eye protection, doors should be locked and windows covered to protect those outside the theatre (Kitching and Edge, 2003).

X-rays

In accordance with the Ionising Radiation Regulations 1999, staff who work with ionising radiation should limit exposure to no more than is reasonably necessary, and should exercise reasonable care while carrying out such work.

Every employee who works with ionising radiation should make full and proper use of the protective equipment provided.

The number of people present in theatre should be limited to those necessary for the procedure and those present must wear lead aprons.

It is advisable for pregnant women to avoid X-rays because radiation may cause foetal abnormalities (Smith, 2000).

Hazards from volatile liquids or gases

avoid the use of flammable solutions with diathermy as there is the risk of ignition.

Anyone affected should leave the theatre immediately.

Soft contact lenses should be removed when methyl methacrylate is being mixed as they are permeable to irritant vapours.

Gas cylinders used in theatre must be clearly labelled with the name or chemical symbol of the gas on the shoulder of the cylinder and the valve.

They should be stored in a cool, well-ventilated room, free from flammable materials. No lubricant of any description should be used on the cylinder valves.

1.2.1.4 Learning Activities

1.2.1.5.1 Case study

You are working as a circulator during surgery when you notice that the surgeon has a hole in his glove. You inform him and he responds that you should not worry about it. What will be your response?

1.2.1.5.2 Field Visit to Theatre

Objectives of the visit

To identify protective gears according to WHO standards

To use protective gears according to WHO standards

Self-Assessment

You are provided with the following questions for self -assessment, attempt them and check your responses

What are three (3) PPEs worn for protection of mucous membranes?

Perioperative Theatre precautions are paramount in ensuring patient safety. Explain five(5) recommended practices for patient safety goals by the Joint Commission that are directly specific to theatre environment?

Describe five(5) specific interventions that you will make to ensure safety in use of a surgical diathermy

What are three (3) interventions you will make to prevent hazards from volatile gases in theatre?

What is the difference between use of sterile gloves and examination gloves?

What is the responsibility of a theatre technologist in ensuring occupation health safety?

Efficient measures in infection prevention and control require that theatre personnel don and doff PPEs correctly. Explain the right procedure for closed gloving Explain the right procedure for closed gloving after donning a sterile gown

Describe the procedure for gowning after a surgical hand scrub

Tools, Equipment, Supplies and Materials

- Medical PPE: Sterile surgical gloves, n95 mask, surgical mask, surgical caps, theatre scrub gown, surgical boots
- Posters on wearing and removal of PPE
- Running water
- Antiseptic solutions
- Hand washing soap
- Hand towels

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RESPONSES

What are three (3) PPEs worn for protection of mucous membranes?

Goggles

Surgical mask

Face shield

Perioperative Theatre precautions are paramount in ensuring patient safety. What are five(5) recommended practices for patient safety goals by the Joint Commission that are directly specific to theatre environment?

Improve the accuracy of patient identification

Improve effectiveness of communication among caregivers

Improve safety of using medications

Reduce the risk of health care–associated infections

Accurately and completely reconcile medications across continuum of care

Reduce the risk of patient harm resulting from falls

Reduce the risk of influenza and pneumococcal disease in institutionalized older adults

Reduce the risk of surgical fires

Implement applicable National Patient Safety Goals and associated requirements by components and practitioner sites

Encourage patients' active involvement in their own care as a patient safety strategy

Prevent health care–associated pressure ulcers (decubitus ulcers)

Identify safety risks inherent in the organization's patient population

Improve recognition and response to changes in a patient's condition

What are five(5) specific interventions that you will make to ensure safety in use of a surgical diathermy?

The patient's skin should be evaluated before and after diathermy use.

The patient should not be in contact with grounded metal objects as the provision of an alternative pathway for the electrical current could result in a burn.

Any metal parts of the operating table that could come into contact with the patient should therefore be padded and jewellery and piercings should be removed before arrival in theatre. The return electrode should be placed over a vascular, muscular area to promote electrical conductivity and to dissipate heat.

Shaving may be necessary as hair at the dispersive site prevents complete plate contact with skin.

The plate should not be placed over bony prominences (this can impede return current), over implanted prostheses (there is a potential for overheating), over scar tissue or over areas distal to tourniquets where adequate tissue perfusion cannot be guaranteed.

What are three (3) interventions you will prevent hazards from volatile gases in theatre?

avoid the use of flammable solutions with diathermy as there is the risk of ignition.

Anyone affected should leave the theatre immediately.

Soft contact lenses should be removed when methyl methacrylate is being mixed as they are permeable to irritant vapours.

Gas cylinders used in theatre must be clearly labelled with the name or chemical symbol of the gas on the shoulder of the cylinder and the valve.

They should be stored in a cool, well-ventilated room, free from flammable materials. No lubricant of any description should be used on the cylinder valves

What is the difference between use of sterile gloves and examination gloves?

Sterile gloves are manufactured to meet the Food and Drugs Administration (FDA) standards for sterilization techniques while examination gloves are not sterilized by the manufacturer. Sterile gloves are used for invasive/ surgical procedures while examination gloves are used for non- sterile procedures

Surgical gloves have a lower acceptable quality level of pinholes (AQL) than examination gloves

Efficient measures in infection prevention and control require that theatre personnel don and doff PPEs correctly. Explain the right procedure for closed gloving after donning a sterile gown

maneuver sterile gloves onto your hands with your hands hidden from view under the gown's cuffs.

Fold the glove wrapper so that the side edges come together at the middle. The gloves are oriented in the wrapper with the fingers up and the cuffed (wrist) part at the bottom edge of the wrapper.

Fold the upper and lower edges of the wrapper inward.

To open the wrapper, grasp the two center edges and open them outward to expose the gloves. To keep the edges from closing up again, evert them slightly; this will remove some of the memory in the folds.

Position the left hand with the palm facing upward, as if you are about to receive an object in your hand. Pick up the glove with your right hand shielded by the gown and place the glove, palm to palm and cuff to cuff, over the left hand. The glove is oriented correctly if the glove fingers point to your elbow.

Working inside the gown cuff, grasp the under edge of the glove cuff between your left thumb and fingers. Using your protected right hand, grasp the upper edge of the glove cuff. The palm of the glove should still be oriented to your palm.

Keep the hidden fingers within 1 inch (2.5 cm) of the outside edge of the knitted cuff, and make sure your thumb is well inside the seam of the cuff.

Pull the glove on. Grasp the left glove cuff and advance your left hand into the glove.

Repeat with the other hand. After gloving, check both hands for any sign of punctures or tears. If a defect is apparent, the circulator removes the glove, and gloving is repeated on that hand.

What is the responsibility of a theatre technologist in ensuring occupation health safety?

Ensure their own safety and health and that of persons who may be affected by their act of commission or omission.

Cooperate with the employer and co-workers in their workplace to achieve a safe and healthy work environment by following OSH policy guidelines and standard operating procedures

Make themselves familiar with the OSH policy guidelines

Use PPE as provided by employer to prevent risk to his/her safety and health

Comply with safety and health Standard Operating Procedures (SOPs)

Report to the supervisor of possible hazards

Report to supervisor on any accidents, injuries, or near misses that arise in the workplace by completing the appropriate form.

If an urgent action is required, the situation must be reported and information forwarded to the appropriate department as soon as possible (24 hours). Report any serious and fatal accidents within 8 hours.

Facilitate the performance of duties given by the employer

Inform the supervisor of any personal condition that may increase vulnerability of occupational hazards

Describe the procedure for gowning after a surgical hand scrub

After drying the hands and arms, grasp the gown just below the neckline and lift it up and away from the table, without touching anything else with bare hands. Remember, the inside surface of the gown faces inward, toward you.

Step away from the table and allow the gown to unfold in front of you. Do not touch the (outside) surface facing away from you.

Being careful not to lower the gown, look for the armholes and place your hands and arms inside the sleeves. Advance your hands, pushing them through to within about 1 inch (2.5 cm) of the knitted cuff edge. Do not allow the fingertips to extend beyond the knitted cuff and sleeve. Avoid raising your arms as you push them through the sleeves. At this point the circulator may secure the neck and inside ties. The circulator assists in securing the gown's back wrap (discussed next). Keep hands well inside the gown cuff and proceed to glove using the *closed* technique

Learning Outcome 2 Verify perioperative Theatre checklist

Introduction to the learning outcome

This learning outcome specifies the content of competencies required to plan for Execution of Perioperative Theatre Safety Precautions. It includes Various Perioperative Theatre checklists and their contents, identification and verification of Perioperative Theatre checklists, record keeping and operation Theatre Practice.

Performance Standard

1. Perioperative checklist is identified according to the workplace unit
2. Perioperative checklist is verified according to Standard Operation Procedures and WHO

Information Sheet

Definition of terms

Checklist- list of required items or actions to be made before proceeding to a procedure

Surgical safety checklist - a tool that is designed to improve the safety of surgical procedures by ensuring that key members of a surgical operation perform key safety checks during the vital phases of perioperative care.

Various Peri-operative Theatre checklists and their contents

According to WHO safe surgery programme, unsafe surgical care can cause substantial harm. Perioperative theatre checklists are a core component of the set standards for theatre safety. Various types of theatre checklists include the WHO surgical safety checklist, checklist for starting the skin preparation and preoperative checklist.

Preoperative checklist

This contains critical elements that must be obtained and verified preoperatively. It is completed by the nurse, usually in the ward or outpatient department, and it accompanies a patient to the operation theatre. Hospitals and other surgical facilities have individual check-in protocols. However, specific details are always verified:

1. Patient identity is meticulously checked- health care provider asks the patient his or her name and verifies this with the patient's unique identifiers, the surgery schedule, and the medical records at hand.
2. Correct procedure, side, and site are validated with the patient, the medical record, the surgical schedule, and the consent form. Preoperative procedures include the surgeon's skin markings on the operative side showing the location of the incision. These are matched with all other information available.
3. Surgical and anesthesia consent forms must be signed according to facility protocol.
4. Resuscitation orders and any other legal documents are checked.
5. Patient allergies must be noted on all medical records, and the patient is asked about allergies again in the holding area.
6. Preoperative medications are documented in the patient's medical and preoperative records. Any medication ordered but not yet given may be administered in the holding area as directed by the surgeon or ACP.
7. Prostheses, including dentures and hearing aids, must be removed before surgery whenever possible. In the event a prosthesis is removed in the holding area, extreme care is taken to protect it from loss or misidentification.

8. Jewelry, including body-piercing jewelry, is removed before anesthesia or any procedure in which electrosurgery is used. Any jewelry removed in the holding area is placed in a container, labeled, and placed in a secure location until it can be safely returned to the patient. A wedding ring may be taped in place.

9. Medical records accompanying the patient are noted. Diagnostic results accompanying the patient, such as radiographs or other imaging studies, are clearly labeled.

Example of perioperative checklist.

Patient's name: _____ Date: _____ Height: _____
Weight: _____
Identification _____ band _____ present: _____

2. Informed consent signed: _____ Special permits signed: _____

3. Surgical site: _____ (Ex: Sterilization): _____

4. History & physical examination report present: _____ Date: _____

5. _____ Laboratory _____ records
present: _____

CBC: _____ Hgb: _____ Urinalysis: _____
Hct: _____

6. Item Present Removed

Natural teeth _____

Dentures; upper, lower, partial _____

Bridge _____

Contact lenses _____

Other prostheses—type: _____

Jewelry:

Wedding band (taped/tied) _____

Rings _____

Earrings: pierced, clip-on _____

Neck chains _____

Any other body piercings _____

Make-up _____

Nail polish _____

7. Clothing

a. Clean patient gown _____

b. Cap _____

c. Sanitary pad, etc. _____

8. Family instructed where to wait?

9. Valuables secured?

10. Blood available? _____ Ordered? _____ Where?

11. Anesthetist medication given:

Type: _____ Time: _____

12. Voided: _____ Amount: _____ Time:
_____ Catheter: _____

13. Mouth care given: _____

14. Vital signs: Temperature: _____ Pulse: _____ Resp:
_____ Blood Pressure: _____

15. Special problems/precautions: (Allergies, deafness, etc.):

16. Area of skin preparation:

17. _____ Date:
_____ Time: _____

Signature: Nurse releasing patient

WHO surgical safety checklist

Research suggests use of a surgical safety checklist reduces morbidity and mortality (Haynes, Weiser, Berry, et al., 2009). It is used prior to induction of anesthesia, before the skin incision is made, and before the patient leaves the OR. It is important to review the patient's record for the following: Correct informed surgical consent, with patient's signature, completed records for health history and physical examination, results of diagnostic studies and allergies (including latex). It is a 19-item checklist that was developed in consultation with various stakeholders with an aim of increasing team work and communication; and decreasing adverse events and errors in surgery.

Surgical Safety Checklist



World Health
Organization

Patient Safety
A World Alliance for Safer Health Care

Before induction of anaesthesia	Before skin incision	Before patient leaves operating room
(with at least nurse and anaesthetist)	(with nurse, anaesthetist and surgeon)	(with nurse, anaesthetist and surgeon)
Has the patient confirmed his/her identity, site, procedure, and consent? ☐ Yes Is the site marked? ☐ Yes ☐ Not applicable Is the anaesthesia machine and medication check complete? ☐ Yes Is the pulse oximeter on the patient and functioning? ☐ Yes Does the patient have a: Known allergy? ☐ No ☐ Yes Difficult airway or aspiration risk? ☐ No ☐ Yes, and equipment/assistance available Risk of >500ml blood loss (7ml/kg in children)? ☐ No ☐ Yes, and two IVs/central access and fluids planned	☐ Confirm all team members have introduced themselves by name and role. ☐ Confirm the patient's name, procedure, and where the incision will be made. Has antibiotic prophylaxis been given within the last 60 minutes? ☐ Yes ☐ Not applicable Anticipated Critical Events To Surgeon: ☐ What are the critical or non-routine steps? ☐ How long will the case take? ☐ What is the anticipated blood loss? To Anaesthetist: ☐ Are there any patient-specific concerns? To Nursing Team: ☐ Has sterility (including indicator results) been confirmed? ☐ Are there equipment issues or any concerns? Is essential imaging displayed? ☐ Yes ☐ Not applicable	Nurse Verbally Confirms: ☐ The name of the procedure ☐ Completion of instrument, sponge and needle counts ☐ Specimen labelling (read specimen labels aloud, including patient name) ☐ Whether there are any equipment problems to be addressed To Surgeon, Anaesthetist and Nurse: ☐ What are the key concerns for recovery and management of this patient?

This checklist is not intended to be comprehensive. Additions and modifications to fit local practice are encouraged.

Revised 1 / 2009

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Redrawn from: https://www.who.int/patientsafety/safesurgery/sssl_checklist_front-thumb.gif

Video link to use of WHO surgical safety checklist: <https://youtu.be/CsNpfMldtyk>
Checklist for starting skin preparation

This should be done before skin incision and it may be done by a theatre technician depending on institution policy. It includes the following:-

Prepare the patient.

Have you checked the patient's record for allergies?

Has the patient been positioned properly?

Has the surgical site been verified?

Has all jewelry been removed?

Has the anesthesia care provider given permission to start the prep?

Are the surgeons present and available to start surgery?

Prepare the supplies.

Note which items are not included in the prep kit.

Are sterile gloves available?

Have the prep solutions been poured?

Is the prep table positioned close to the patient?

Is a receptacle at hand for soiled prep sponges?

Do you have adequate light on the prep area?

Prepare yourself.

Do you have a plan?

Do you know the exact boundaries of the prep area?

Is your clothing contained so that it does not touch the prep area? (A loose warm-up jacket or baggy sleeves may drag across the prep area.)

Identification and verification of Perioperative Theatre checklists

A perioperative theatre technologist should collaborate with the nurse in check listing the patient for surgery as provided for in the scope of practice and medical legal ethics. In the operation room, the perioperative checklist that is escorting the patient to theatre should be counter-checked by the receiving personnel. Following admission in theatre, the WHO surgical safety checklist and the skin preparation checklist should be checked before the operation is started. Other checklists should be reviewed according to institutional policies.

Record keeping and operation Theatre Practice

Documentation is the means of making a permanent legal record of the patient's interaction with health care providers and services. It is a way for health professionals to communicate patient procedures, diagnoses, treatments, condition, and recommendations for care. Separate forms are required for many different types of documentation, but the method of documentation is consistent among health care facilities. The importance of correct documentation cannot be overstated. A mistake in documentation can lead to serious medical errors. Different registers used in operation theatre include the following:-

Instrument register- It mentions the total number of instrument sets used in a single day in the various OTs to help keep track of lost instruments. □

Linen stock register- It details the linen type and quantity that the OT has.

Complaint/incident register- it records all the unusual incidences and complaints in the OT.

Microbiology register- it maintains a record of microbiology reports for various samples taken.

Water analysis register- it maintains the results for microbiological tests on water samples from various areas of the OT like scrub room and sterilization room.

Daily register- It details the infection control activities of OT as follows: Name of the concerned cleaning staff on duty, assessment of temperature control, samples sent from the OT for microbiological tests, biomedical waste management, fumigation details of the OT with the name of chemical reagent used.

Operation register- it records all the surgical procedures done in the operation room. it records all the information and particulars of the patient to include the patient's full name, age, sex, registration number, ward in which the patient is admitted, full description of the surgery performed, name of anesthesiologist, name of the operating surgeon and assistant, anesthetic agent used for anesthesia, name of the scrub nurses, name of the OT assistant, operation theater table on which the patient was operated, whether the procedure was a major or minor surgery and the duration of the operation.

Sterilization registers- it includes the following details: number of instrument sets along with the method of sterilization, the total time taken for sterilization along with its starting and ending time, the staff responsible for sterilization.

Temperature and humidity control register- it includes the checked temperature whose recording is done every two hours.

Central sterile supply department (CSSD) register- it mentions the details of the quantity of linen sent for sterilization and received after the sterilization procedure is over.

Link to further details on medical records in operation theatre:
https://www.researchgate.net/publication/325855465_Operation_Theater_List_and_Record_Maintenance

The following general guidelines should be followed for health care documentation (to include the perioperative theatre practice documentation):

1. Every document must contain the patient's unique identifiers, including patient name, hospital or identity card number, and other information required specifically for that health facility.
2. The date must be accurate. Never predate or postdate a document. Always document the correct time.
3. If you make an error in handwritten documentation, make a single line through the part that is incorrect and write in the correct information. Initial the change. Never use opaque liquids or tapes to blank out the error—it must remain in the record.

If the documentation is performed in writing, make sure it is legible.

Documents must be kept clean and dry to prevent smearing. It is good practice to document away from areas where spills or splashes can occur.

Learning Activities

Case study

You are participating in check-listing of a patient during the sign-out time when you realize that an operation has been performed on the wrong patient. What will be your immediate response?

Field Visit to Theatre

Objectives of the visit

To identify the perioperative checklist according to the workplace unit

To verify the perioperative checklist according to Standard Operation Procedures and WHO Self-Assessment

You are provided with the following questions for self -assessment, attempt them and check your responses

What is a surgical safety checklist?

What are five specific details that must be verified in theatre checklists?

Documentation and record keeping serves many functions in operation theatre to include fulfilling the medical-legal requirements. What details should be included in an operation register?

What are five general guidelines that should be followed for health care documentation?

What should be confirmed on a theatre safety checklist during sign-in?

What are the components of a safety checklist during sign out?

Skin preparation checklist is a requirement before initiation of an operation. What are the components of patient preparation in this checklist?

A daily operation theatre register details the infection control activities of OT. What are common components of this register?

Tools, Equipment, Supplies and Materials

Sample surgical safety checklist

Institutional preoperation checklist

skin preparation checklists

Sterile surgical gloves

Theatre records to include operation registers, daily register, sterilization register, temperature and humidity control register, microbiology register, water analysis and culture register, instrument register, central sterile supply department register(CSSD), linen stock register, complaints register.

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RESPONSES

What is a surgical safety checklist?

A surgical safety checklist is a tool that is designed to improve the safety of surgical procedures by ensuring that key members of a surgical operation perform key safety checks during the vital phases of perioperative care.

What are five specific details that must be verified in theatre checklists?

Patient identity

Correct procedure

Surgical and anesthesia consent forms- signed according to facility protocol.

Resuscitation orders and any other legal documents

Patient allergies

Preoperative medications

Prostheses

Jewelry

Medical records accompanying the patient

Documentation and record keeping serves many functions in operation theatre to include fulfilling the medical-legal requirements. What details should be included in an operation register?

all the surgical procedures done in the operation room.

all the information and particulars of the patient to include the patient's full name, age, sex, registration number, ward in which the patient is admitted

full description of the surgery performed

name of anesthesiologist

name of the operating surgeon and assistant

anesthetic agent used for anesthesia

name of the scrub nurses and the OT assistant

operation theater table on which the patient was operated

whether the procedure was a major or minor surgery and the duration of the operation.

What are five general guidelines that should be followed for health care documentation?

Every document must contain the patient's unique identifiers, including patient name, hospital or identity card number, and other information required specifically for that health facility.

The date must be accurate. Never predate or postdate a document. Always document the correct time.

If you make an error in handwritten documentation, make a single line through the part that is incorrect and write in the correct information. Initial the change. Never use opaque liquids or tapes to blank out the error—it must remain in the record.

If the documentation is performed in writing, make sure it is legible.

Documents must be kept clean and dry to prevent smearing.

What should be confirmed on a theatre safety checklist during sign-in?

Confirm the patient identity, procedure and site to be operated

Patient allergies

Difficulty airway or aspiration risk

Risk of blood loss

Anesthesia safety completed

Briefing all members of surgical team on care of patient or any concerns

What are the components of a safety checklist during sign out?

Confirm name of operative procedure

Instrument count

Specimen identified and labelled

Any equipment problem to be addressed

Skin preparation checklist is a requirement before initiation of an operation. What are the components of patient preparation in this checklist?

Have you checked the patient's record for allergies?

Has the patient been positioned properly?

Has the surgical site been verified?

Has all jewelry been removed?

Has the anesthesia care provider given permission to start the prep?

Are the surgeons present and available to start surgery?

A daily operation theatre register details the infection control activities of OT. What are common components of this register?

Name of the concerned cleaning staff on duty

assessment of temperature control

samples sent from the OT for microbiological tests

biomedical waste management

fumigation details of the OT with the name of chemical reagent used

Learning Outcome 3 Care for patient and theatre team

Introduction to the learning outcome

This learning outcome specifies the content of competencies required to plan for Execution of Perioperative Theatre Safety Precautions. It includes Identification of patients, Operation Theatre team, Team work, Transferring patient to the operation table, Basic nursing skills, Basic life support, Positioners and Positioning of patient on operation table, Making drills and evacuation and Operation Theatre Practice

Performance Standard

1. Patient is identified according to procedure to be performed; IP. No, file and identification band
2. Completeness of theatre team is checked according to WHO standards
3. Patient is transferred to the operation table according to SOPs
4. Patient is positioned according to procedure to be performed
5. Positioners are strapped onto the patient as per SOPs
6. Patient is transferred to PT trolley as per SOPs
7. Drills and Evacuation procedure is carried out according to OSH Act

1.2.3.3 Information Sheet

Definition of terms

Team work- combined group work directed at efficiently performing a given task

Basic life support-Basic life support (BLS) includes recognition of signs of sudden cardiac arrest (SCA), heart attack, stroke, and foreign-body airway obstruction (FBAO); cardiopulmonary resuscitation (CPR); and defibrillation with an automated external defibrillator (American Heart Association (AHA), 2010).

Positioners- a device used by the surgical team members to align the patient's body for a given surgical procedure

Fire drill- method of practicing how a building would be evacuated in the event of a fire or other emergencies

Identification of patients

Use of a standard process to verify patient identification, the surgical procedure, and the surgical site is imperative to maximize patient safety (World Health Organization, 2008). This allows for prompt intervention if any discrepancies are identified.

The identity of the patient is verified before transporting and before any procedure. Patient identification is a critical issue in health care. The perioperative theatre technologist is responsible for patient identification according to facility policy and the mutual guidelines agreed on by all professional surgical organizations. No patient should be transported and no procedure should be initiated until the protocol for identification has been completed, even if the patient is known to the health care staff.

All patients are identified in at least three ways. The patient's wrist or ankle band is imprinted with the patient's name and other unique identifiers such as birth date and hospital number. If a scan or imprint card system is used, the patient's identification card is used to process all paperwork and matches the patient's identification bracelet. This card must be firmly attached to the chart during transport and must remain with the chart until the patient returns to his or her hospital unit. The patient's chart must accompany the patient whenever the individual is transported from the unit.

To validate the patient's identity, follow these guidelines:

1. Examine the patient's identity band. Compare both the name and the number with those on the patient's chart.
2. Ask the patient to state his or her full name and date of birth. Do not call the patient by name before asking the patient to state his or her name.
3. Ask the patient to tell you what procedure he or she is undergoing and to point to the side on which the surgery will take place.
4. If the patient does not speak English, or seems to have difficulty understanding, you must seek assistance from an interpreter. This information can be determined ahead of time so that an appropriate interpreter is available.
5. Remember that patients may be anxious or worried before surgery and might answer closed-ended questions indiscriminately. It is necessary to question the patient without giving the answer.
6. Always check the chart, the identification band, and hospital ID number for each patient.

Operation Theatre Team

The surgical team consists of the patient, the anesthesiologist or anesthetist, the surgeon, nurses, and the surgical technologists (or assistants). They have unique roles during an operation as outlined below:-

The anesthesiologist or anesthetist administers the anesthetic agent and monitors the patient's physical status throughout the surgery.

Surgical assistants scrub and perform the surgery. The person in the scrub role, either a nurse or a theatre technologist, provides sterile instruments and supplies to the surgeon during the procedure.

The circulating nurse coordinates the care of the patient in the Operation room (OR). Care provided by the circulating nurse includes assisting with patient positioning, preparing the patient's skin for surgery, managing surgical specimens, anticipating the needs of the surgical team, and documenting intraoperative events.

The surgeon- participates as both a sterile and nonsterile team member. At the start of the case, the surgeon performs sterile techniques to place the trocars. He or she then breaks scrub

to operate from the surgeon's console. The surgeon returns to the sterile field (after scrubbing, regowning, and gloving) near the close of the procedure to remove the trocars and close the incisions. The surgeon directs the flow of the procedure and is responsible for coordinating the activities of everyone on the team.

The patient

Team work

A team is a group of people who come together to reach a common goal or set of goals. The surgical team is only one type of team that plans and implements patient care in the operating room. In some large hospitals, certain personnel work within a surgical specialty, such as cardiology or orthopedics. In this type of structure, surgical technologists work with their peers to design instrument sets, order equipment, and update the surgeons' procedural changes. The team may or may not have a team leader.

The surgical team includes the surgeons, anesthesia provider, assistants, surgical technologist, and registered nurse. They all work together on a single procedure. Communication usually is focused, task oriented, and at times intense.

Characteristics of good teamwork

Good teamwork is the result of healthy relationships within the team. This does not mean that conflicts do not arise. Conflict in groups is normal, because people have different ideas, problem-solving skills, values, and beliefs. The qualities of a good team reflect how conflict is managed and include:-

Yielding - This does not mean giving up one's values or beliefs. It means accepting the fact that others have valid points of view and conceding when one has made incorrect assumptions or conclusions.

Change - One of the purposes of a team is to adjust to a changing environment, such as unfolding events during a surgical procedure, a change in instrumentation, or new responsibilities. Team members must identify new tasks or procedures and implement them with as little disruption as possible.

Politeness - This manner in which people speak to and behave toward each other. The attributes of acceptable behavior include respect, gratitude, and acceptance.

Collaboration- This is working together for a common purpose. In the operating room, personnel contribute their skills, time, and energy to the care of the surgical patient.

Transferring patient to the operation table

Before the patient is transferred to the operation room, the following should be verified: Patient identity, procedure to be performed and a signed consent form. The following factors determine whether a patient can ambulate to the operation bed, use a wheelchair or a stretcher: Presence of invasive devices, state of the patient, planned procedure and the institutional policy.

Stretchers should have side rails and lock mechanisms and the head should elevate to alter the patients comfort and position. The side rails are kept up and safety strap secured during transport. The staff member who is close to the proximity of the patient's airway should push the stretcher. In collaboration with a nurse, you should assess the patient's condition to determine whether special equipment like an ICU bed are required during transport.

Once sufficient staff members are present, the stretcher/bed is wheeled close to the operation table, the wheels of both the operation bed and the stretcher/bed locked and the side rail that is proximal to the operation bed lowered. Invasive lines and devices are made visible and kept free from entanglement. Some team members stand at the far end of the operation table; others on the proximal end and use appropriate body mechanisms to prevent injury to self during transferring of the patient. A roller or a transfer sheet is used to transfer a patient if they are not mobile and invasive devices supported in place to prevent entanglement.

Basic Nursing skills

It is important to be competent in the following procedures because they are vital during patient positioning:

Observation of vital signs

These include Blood Pressure; Temperature; Pulse and Respiration. They help in determining the most appropriate positions for a patient. For example, a patient with hypotension or bradypnea may not be positioned in reverse Trendelenburg position.

Bed making procedures

Video link to bed making procedures: <https://notesread.com/bed-making-procedure-in-nursing/>

Body mechanics in bed making

When bending, bend your knees, not your back in order to keep the center of gravity above and close to the base of support and help prevent fatigue

Face your entire body in the direction that you are moving to avoid twisting in order to prevent strain or injury

Organize your work. Conserve your steps by making a few trips around the bed as possible

Appliances used in bed making

Mackintosh- Water proof material used to protect the bottom sheets, pillows and mattress from getting wet

Bed Rest/Back Rest- Helps the patient sit in upright position. May be attached to or separate from the bed and can be adjusted to different degrees

Fracture Boards- They are boards (wooden) placed under the mattress to provide a firm rigid foundation to the bed and prevent the mattress from sagging. Aid in immobilization on affected part of the body.

Bed Cradle- Appliance used to keep the weight of the linen off the patient e.g. in patient with burns to protect linen from touching the area with wound

Sand Bags- Small bags made of impermeable material filled with sand. They are used to support patients legs to prevent foot drop. May also be used to immobilize a fractured limb

Bed Blocks- Made of wood and used to raise the bed on one side e.g. to supply blood to vital organs like the brain

Air Rings -Rubber ring placed beneath the patient's buttocks to keep them from direct contact with the bed. Relieve pressure hence prevent pressure sores.

Bed Tables- Specially made tables which can be drawn up in front of the patient and may be used during meals or for leaning slightly forward in case the patient has difficulty in breathing

Hot Water Bottles- Bottles of rubber used to give warmth to the patient. They may also be used to relieve pain

Foot Boards- Used to secure and support the patients foot

Water Beds- They are just like mattress though have a place to fill with hot water

Air Beds- These are mattresses that are filled with air

Ripple Mattresses- Have segments and uses electricity to put air pressure. Used to relieve pressure

Basic life support

Knowledge and certification in Cardiopulmonary resuscitation (CPR) is paramount in ensuring that health care staff are able to respond in case of a cardiac or respiratory arrest. The goal of CPR is to support and restore oxygenation, ventilation, and circulation. Restoration of intact neurological function accompanies this process. Return of spontaneous circulation is accomplished with basic life support (BLS) or advanced cardiac life support (ACLS) measures. A perioperative theatre technologist should possess knowledge and skills in BLS as learnt in the unit of competency “ Provide First Aid Services”. He or she should also be able to assess for signs of Cardiopulmonary arrest which may include nausea, shortness of breath, chest pain or pressure, pain radiating from the jaw, neck, or Shoulder and cardiac dysrhythmias.

Positioners and Positioning of patient on operation table

The surgical patient is positioned on the operating table for a specific surgical procedure:

To allow optimum access to the operative site

To reduce adverse physiological effects and prevent injury during surgery

To permit optimum access for anesthesia care (includes access to venous and arterial sites, airway, and monitoring sites)

The following elements are important in the safe positioning of patients:

Positioning - Each patient is unique and has specific considerations, such as age, joint mobility, and disease. Because of the high risk of serious and permanent injury, the surgical team must be guided and directed in the positioning process. The anesthesia care provider, surgeon, and circulator draw this direction from their knowledge of the patient's status.

Planning- All necessary equipment must be assembled ahead of time. Pads, positioning devices, table accessories, and transfer devices must be on hand before positioning begins. Bariatric/ obese table extensions or special tables must be on hand before the patient arrives in the surgical suite. Adequate personnel must be available to complete the task safely.

Teamwork- Teamwork is needed to create smooth, step by-step coordination.

Theatre technologists have specific responsibilities during patient positioning as outlined below:

Understand common positions and the surgeries for which these positions are used

Know ahead of time the position that will be used for each assigned surgical procedure

Proactively prevent accident and injury during positioning

Question any aspect of the patient's position that appears to have risk potential

Remain alert and focused on patient safety

Communicate clearly with other members of the team

Principles of safe positioning

All equipment needed for positioning must be assembled and prepared for use before the patient is brought into the room.

Adequate personnel must be available to assist before positioning begins. Do not risk the patient's safety because of a crowded surgical schedule.

Before positioning begins, all team members should be familiar with the position and each person must understand his or her role in positioning

The patient should not be moved except on the instructions of the person directing the move. Although everyone involved is responsible for the patient's safety, one person—usually the surgeon, anesthesia care provider, or surgical assistant—guides and directs the others.

In general, the anesthesia care provider must give permission before any change is made in a patient's position because these can cause physiological alterations such as a drop in blood pressure.

Always check equipment before using it. Tighten the locking devices of all weight-bearing accessories.

Make sure the table is locked securely in position and do not assume that any accessory equipment is in working order.

Move slowly when positioning the patient.

Always move the body within its normal range of motion. This requires knowledge of joint types and anatomy.

When positioning an unconscious, sedated, or weak patient, make certain that you have complete control of the part you are moving before you begin.

Surgical positions that may be adopted in relation to the surgery to be performed include:-

Horizontal recumbent/ supine/ dorsal

The patient is placed flat on back with legs extended or slightly flexed. A bath blanket is placed over patient lengthwise. Pillows should be placed under the neck to prevent neck hyperextension. Arms should be placed (pronated) along the trunk. Extra pillows should be placed at the pelvis to prevent hyper rotation

Dorsal recumbent

Patient is placed flat on back with one pillow under head; the knees flexed and separated and feet flat on bed. A blanket is arranged to cover extremities and expose perineum.

Prone position

Patient lies on his/her abdomen. The head is turned to either side for comfort. Pillow is placed under the lower chest, pelvis and ankles for comfort. Arms may be placed on the side or flexed near the head

Lateral/side lying

Patient is placed on the side with the head supported on low pillow. A pillow is used to support the body anteriorly and another one is tucked at the waist from the back to support the patient in the position. Legs are flexed forwards and pillow a placed lengthwise between the legs to separate them and support the top leg

Sims/recovery/semi prone position

The patient is placed on left side somewhat obliquely across the bed with buttocks to edge of mattress. The body is inclined forward, the left arm drawn back under patient and the right

arm placed in front. The knees are flexed upon the body—the right more than the left. A pillow is placed under the head

Semi – Fowlers Position

The bed is tilted to an angle of 15 – 30 degrees

High Fowlers Position

Used when the head of the bed needs to be elevated as high as possible. The upper half of the patient's body is between 60 degrees and 90 degrees in relation to the lower half of their body. The legs of the patient may be straight or bent.

Trendelenburg position

Client lies in supine position on an adjustable bed or operating table. S/he is then secured and supported using straps on the table before adjusting the bed/table. The bed/table is then tilted to lower the head at an angle of 45 degrees below the horizontal plane.

Reverse trendelenburg

The reverse trendelenburg position is used to expose the prostate and upper abdominal region during surgery. This position also helps respiration for overweight and obese individuals.

Dorsal lithotomy

The patient is placed in supine position on adjustable bed/ examination table/operating table. Client's buttocks are placed as close as possible to the edge of the bed/table. Both knees are flexed simultaneously and supported on stirrups to prevent injury when each leg is lifted singly

Knee – chest/genu-pectoral position

Client is positioned in prone position. The client then assisted to kneel on the bed raising hips to the air and resting her weight on her chest and knees. The head is turned to one side, supported on a pillow and arms flexed at the elbows extending, then to the bed in front of her.

Jack knife or kroaske or bozeman

Patient is placed on a prone position with the hips directly over the band of the examining table. The table is tipped with the head lower than the hips, the foot part of the table lowered so that the patients feet are below the level of his head. A pillow is placed under the pelvis and abdomen to relieve the strain.



Fig. 14.1. Supine position with arms in abduction.

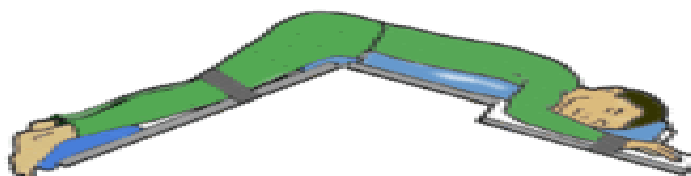
Supine position retrieved from: https://www.europeanmedical.info/surgical-applications/images/8917_65_230-ureteral-stent-placement-procedure.jpg



Lithotomy position retrieved from: https://www.europeanmedical.info/surgical-applications/images/8917_65_233-lithotomy-position.jpg



Prone position retrieved from: https://www.europeanmedical.info/surgical-applications/images/8917_65_242-arm-abduction-table-positioning.jpg



Jackknife position retrieved from:
https://upload.wikimedia.org/wikipedia/commons/thumb/f/f6/Jack_knife_position.gif/220px-Jack_knife_position.gif



Fowlers position retrieved from: <https://www.steris.com/-/media/images/knowledge-center/surgical-equipment/complete-guide-to-patient-positioning/fowlers.ashx?h=250&w=242&hash=1F9F866B55B958408E6029C2D00198EC>



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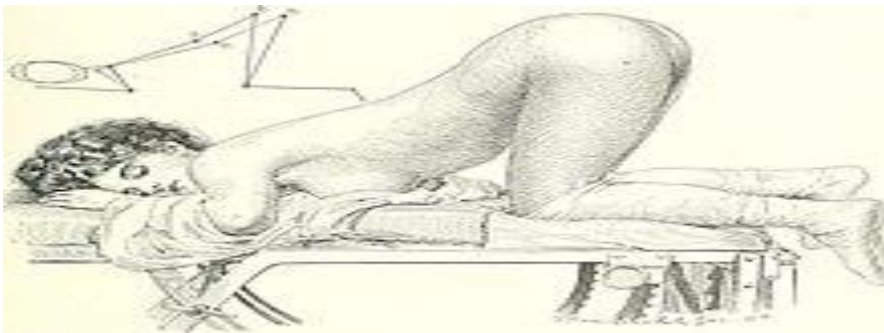


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[positioning/lateral.ashx?h=160&w=242&hash=14072CF871DD7C5185748CBF872009ED](#)



Trendelenburg position retrieved from: <https://www.steris.com/-/media/images/knowledge-center/surgical-equipment/the-ultimate-guide-to-the-trendelenburg-position/trendelenburg-position-picture.ashx?h=200&w=250&hash=1F3BE469A7D7EF784B9BE3C4BEBE228A>



Knee chest position link adress:
https://upload.wikimedia.org/wikipedia/commons/thumb/6/60/Diseases_of_the_kidneys%2C_ureters_and_bladder%2C_with_special_reference_to_the_diseases_of_women_%281922%29_%2814763493624%29.jpg/220px-



Reverse trendelenburg position image adress: <https://www.steris.com/-/media/images/knowledge-center/surgical-equipment/the-ultimate-guide-to-the-trendelenburg-position/reverse-trendelenburg-position-picture.ashx?h=200&w=250&hash=01491BB097670B7329D6CE217E6E854F>

Making drills and evacuation

Fire drills are held regularly in all health care facilities. Staff training on fire includes emergency response, the location of fire extinguishers and fire escape routes, and how to activate the fire alert system. Most fire extinguishers used in the operating room are water-based, carbon dioxide, or dry powder. Carbon dioxide is the preferred type for operating room fires.

During fire extinguisher training, employees and students are asked to remember the acronym PASS:

Pull the ring from the handle.

Aim the nozzle at the base of the fire.

Squeeze the handle.

Sweep the fire with tank contents

Fire in the Operating Room

Patient fire

Patient fire is a devastating event. Approximately 21% of patient fires occur in the airway, 44% on the face, 8% inside the patient, and 26% on the skin. It takes only moments for a flash fire to engulf the patient. To stop the progression of the fire, the triangle of fire must be broken. This means that one or more components (**fuel, oxygen, or source of ignition**) must be removed from the fire.

During a patient fire, time is critical. Three steps are immediately taken to protect the patient and stop the fire:

1. Shut off the flow of all gases to the patient's airway.
2. Remove any burning objects from the surgical site.
3. Assess the patient for injury and respond appropriately. The anesthesia care provider reduces the flow of oxygen in the event of fire around the airway. At the same time, burning objects are removed from the field as safely as possible. The surgical technologist must stand by for direction from the surgeon and other staff members in the room. Patient fires usually can be contained when one of the elements of the fire has been removed. The next phase of the emergency focuses on the patient's injuries.

Structural Fire

If the fire extends beyond the immediate patient area, the surgical team must activate the hospital evacuation plan. This plan is based on four immediate actions, which are easily remembered by the acronym RACE:

Rescue patients in the immediate area of the fire.

Alert other people to the fire so that they can assist in patient removal and response. Activate the fire alert system.

Contain the fire. Shut all doors to slow the spread of smoke and flame. Always shut off the zone valves controlling inline gases to the room.

Evacuate personnel in the areas around the fire.

Learning Activities

Case study

You are assisting in monitoring a patient during the post anesthesia phase. Once the patient has recovered from effects of anesthesia and you are planning to inform the perioperative

nurse to conduct her assessment in order to plan for discharge to the ward, you realise that the patient has become unresponsive. What will you do?

Field Visit to Theatre

Objectives of the visit

Patient is identified according to procedure to be performed; IP. No, file and identification band

Completeness of theatre team is checked according to WHO standards

Patient is transferred to the operation table according to SOPs

Patient is positioned according to procedure to be performed

Positioners are strapped onto the patient as per SOPs

Patient is transferred to PT trolley as per SOPs

Drills and Evacuation procedure is carried out according to OSH Act

Self-Assessment

You are provided with the following questions for self -assessment, attempt them and check your responses

What are two ways of identifying a patient correctly in surgical theatre?

What steps will you follow to verify the identity of a patient?

The operation theatre team is composed of different members. What are the roles of the following members?

Surgical assistants

Scrub nurse

What three (3) characteristics that define a good team?

Explain the steps will you follow to safely transfer a patient to the operation table?

What are two goals of correct patient positioning in the operation theatre?

What are your four (4) responsibilities as a theatre technician during patient positioning?

Describe three positions that may be used during a surgical operation

Tools, Equipment, Supplies and Materials

Ideal operation theatre bed

Patient positioners

Bed appliances

Emergency trolley

Fire extinguishers for different fire types

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RESPONSES

What are two ways of identifying a patient correctly in surgical theatre?

The patient's wrist or ankle band is imprinted with the patient's name and other unique identifiers such as birth date and hospital number.

The patient's identification card

The patient's charts

What steps will you follow to verify the identity of a patient?

Examine the patient's identity band. Compare both the name and the number with those on the patient's chart.

Ask the patient to state his or her full name and date of birth. Do not call the patient by name before asking the patient to state his or her name.

Ask the patient to tell you what procedure he or she is undergoing and to point to the side on which the surgery will take place.

If the patient does not speak English, or seems to have difficulty understanding, you must seek assistance from an interpreter. This information can be determined ahead of time so that an appropriate interpreter is available.

Remember that patients may be anxious or worried before surgery and might answer closed-ended questions indiscriminately. It is necessary to question the patient without giving the answer.

The operation theatre team is composed of different members. What are the roles of the following members?

Surgical assistants scrub and perform the surgery. The person in the scrub role, either a nurse or a theatre technologist, provides sterile instruments and supplies to the surgeon during the procedure.

The circulating nurse coordinates the care of the patient in the Operation room (OR). Care provided by the circulating nurse includes assisting with patient positioning, preparing the patient's skin for surgery, managing surgical specimens, anticipating the needs of the surgical team, and documenting intraoperative events.

What three (3) characteristics that define a good team?

Yielding - It means accepting the fact that others have valid points of view and conceding when one has made incorrect assumptions or conclusions.

Change - Team members must identify new tasks or procedures and implement them with as little disruption as possible.

Politeness - This manner in which people speak to and behave toward each other. The attributes of acceptable behavior include respect, gratitude, and acceptance.

Collaboration- This is working together for a common purpose. In the operating room, personnel contribute their skills, time, and energy to the care of the surgical patient.

Explain the steps will you follow to safely transfer a patient to the operation table?

Verify the patient identity, procedure to be performed and a signed consent form. Assess patient's mobility status and institutional policy to determine how to move the patient

Ensure presence of sufficient staff members

Wheel the stretcher/bed is wheeled close to the operation table, the wheels of both the operation bed and the stretcher/bed locked and the side rail that is proximal to the operation bed lowered. Invasive lines and devices are made visible and kept free from entanglement. Some team members stand at the far end of the operation table; others on the proximal end and use appropriate body mechanisms to prevent injury to self during transferring of the patient.

A roller or a transfer sheet is used to transfer a patient if they are not mobile and invasive devices supported in place to prevent entanglement.

What are two goals of correct patient positioning in the operation theatre?

To allow optimum access to the operative site

To reduce adverse physiological effects and prevent injury during surgery

To permit optimum access for anesthesia care (includes access to venous and arterial sites, airway, and monitoring sites)

What are your four (4) responsibilities as a theatre technician during patient positioning?

Understand common positions and the surgeries for which these positions are used

Know ahead of time the position that will be used for each assigned surgical procedure

Proactively prevent accident and injury during positioning

Question any aspect of the patient's position that appears to have risk potential

Remain alert and focused on patient safety

Communicate clearly with other members of the team

Describe three positions that may be used during a surgical operation

Dorsal recumbent- Patient is placed flat on back with one pillow under head; the knees flexed and separated and feet flat on bed. A blanket is arranged to cover extremities and expose perineum.

Prone position- Patient lies on his/her abdomen. The head is turned to either side for comfort. Pillow is placed under the lower chest, pelvis and ankles for comfort. Arms may be placed on the side or flexed near the head

Lateral/side lying- Patient is placed on the side with the head supported on low pillow. A pillow is used to support the body anteriorly and another one is tucked at the waist from the back to support the patient in the position. Legs are flexed forwards and pillow is placed lengthwise between the legs to separate them and support the top leg

Sims/recovery/semi prone position- The patient is placed on left side somewhat obliquely across the bed with buttocks to edge of mattress. The body is inclined forward, the left arm drawn back under patient and the right arm placed in front. The knees are flexed upon the body—the right more than the left. A pillow is placed under the head

Semi – Fowlers Position- The bed is tilted to an angle of 15 – 30 degrees

High Fowlers Position -Used when the head of the bed needs to be elevated as high as possible. The upper half of the patient's body is between 60 degrees and 90 degrees in relation to the lower half of their body. The legs of the patient may be straight or bent.

Trendelenburg position- Client lies in supine position on an adjustable bed or operating table. S/he is then secured and supported using straps on the table before adjusting the bed/table. The bed/table is then tilted to lower the head at an angle of 45 degrees below the horizontal plane.

Reverse trendelenburg- The reverse trendelenburg position is used to expose the prostate and upper abdominal region during surgery. This position also helps respiration for overweight and obese individuals.

Dorsal lithotomy- The patient is placed in supine position on adjustable bed/ examination table/operating table. Client's buttocks are placed as close as possible to the edge of the bed/table. Both knees are flexed simultaneously and supported on stirrups to prevent injury when each leg is lifted singly

Knee – chest/genu-pectoral position- Client is positioned in prone position. The client then assisted to kneel on the bed raising hips to the air and resting her weight on her chest and knees. The head is turned to one side, supported on a pillow and arms flexed at the elbows extending, then to the bed in front of her.

Jack knife or kroaske or bozeman- Patient is placed on a prone position with the hips directly over the band of the examining table. The table is tipped with the head lower than the hips, the foot part of the table lowered so that the patients feet are below the level of his head. A pillow is placed under the pelvis and abdomen to relieve the strain.

Learning Outcome 4 Handle instruments, apparatus and consumables

Introduction to the learning outcome

This learning outcome specifies the content of competencies required to plan for Execution of Perioperative Theatre Safety Precautions. It includes Identification of Operation Theatre Instrument, apparatus and consumables; Operation Theatre Instrument, apparatus and consumables handling techniques; Verification and recording of Operation Theatre Instrument, apparatus and consumables; Disposal of swabs and sharps; Environmental health; Operation Theatre Practice

1.2.4.2 Performance Standard

1. instruments and consumables are identified according to the procedure to be carried out
2. Instruments, swabs and sharps are counted before and after the procedure according to WHO standards
3. Instruments, swabs, sharps and consumables are confirmed and recorded according to the procedure performed
4. Swabs and sharps are discarded according to WHO standards and environmental protection regulation and SOPs

1.2.4.3 Information Sheet

Definition of terms

Swabs- porous materials used to absorb body fluids such as blood or pus from body tissues

Theatre instruments- devices used to perform certain theatre activities such as cutting, scooping, tissue scrapping, coagulation, holding, grasping or even suturing

Identification of Operation Theatre Instrument apparatus and consumables

Expertise in surgical instrumentation is among the most important roles of the surgical technologist. Familiarity with instruments and their names (sometimes different from one region to another, and even among health facilities) is also one of the most difficult learning curves for students in surgery. The most useful way to learn the names and use of instruments is to study them in association with the surgeries they are used in rather than in isolation, such as in a catalog. Most surgical instruments can be classified into these 4 basic types:

Cutting and Dissecting – These instruments usually have sharp edges or tips to cut through skin, tissue and suture material. Surgeons need to cut and dissect tissue to explore irregular growths and to remove dangerous or damaged tissue. These instruments have single or double razor-sharp edges or blades. OR personnel need to be very careful to avoid injuries, and regularly inspect these instruments before using, for re-sharpening or replacement. Cutting instruments include the following:-

Surgical Blade/scapel/ “the knife”

It is used whenever razor sharp cutting is required for tissue dissection. It is detachable from the knife handle, although single-use scalpels are available and these do not detach. They are numbered according to their size and shape and this is known as the Bard-Parker system according to the company that first manufactured them. Each specific number handle is designed for only certain blades and the numbering system is consistent across all manufacturers. The number of the blade corresponds to a specific size and shape.

Types of blades

Blade Number	Picture	Description
› No. 10		✓ A blade with a curved belly. ✓ The belly being the sharp edge. ✓ Used in making skin incisions.
› No. 11		✓ A blade with a straight and an angled edge with a pointed end. ✓ Used in making stab incisions to insert drains.
› No. 12		✓ A blade which is crescent shaped. ✓ The inner side of crescent is sharp. ✓ Used in removing sutures, parotid and cleft palate surgeries.

Image address: <https://epomedicine.com/wpcontent/uploads/2020/07/blade-2.png>

Blade Number	Picture	Description
› No. 15		✓ A blade with small cutting edge. ✓ Used in making precise and short incisions. ✓ Similar to No. 10 but smaller
› No. 20		✓ A blade with a curved belly. ✓ The belly being the sharp edge. ✓ Used in making skin incisions. ✓ Similar to No. 10 but larger.
› No. 21		✓ A blade with a curved belly. ✓ The belly being the sharp edge. ✓ Used in making skin incisions. ✓ Similar to No. 10 but larger. ✓ Larger than No. 20.

Image address:

<https://epomedicine.com/wp-content/uploads/2020/07/blade-3.png>

Blade Number	Picture	Description
› No.22		✓ A blade with small cutting edge. ✓ Used in making skin incisions. ✓ Similar to No. 10 but larger. ✓ Larger than No. 21
› No. 23		✓ A blade with a curved cutting edge. ✓ The opposite side is blunt. ✓ A leaf shaped blade. ✓ Used in making long incisions.
› No. 26		✓ A blade with a straight cutting edge. ✓ The opposite side is straight and blunt with a downward angle. (Note the area circled)

Image address:

<https://epomedicine.com/wp-content/uploads/2020/07/blade-4.png>

Scissors

High quality surgical scissors are distinguished by the sharpness of the cutting edges, balance, metal composition, and design of the point. Scissors with extremely sharp points are intended to sever very small points of tissue during dissection. The blades of the high quality scissor are coated with tungsten or other hardened alloy to maintain sharpness.

Types of surgical scissors

Small, sharp-tipped scissors, such as iris scissors, are used for extremely fine dissection in plastic surgery.

Castroviejo scissors are commonly used in microsurgery.

Round-tipped, light dissecting scissors, such as Metzenbaum scissors, are used extensively on delicate tissue in general surgery. They are the most commonly used scissors in general surgery

Fibrous connective tissue requires heavier scissors, such as the curved Mayo scissors.

Straight Mayo scissors are used for cutting suture. NB: tissue scissors must never be used to cut suture because it dulls them quickly.

Wire-cutting scissors are used for stainless steel and other metal suture materials.



Image address to types of surgical scissors: <https://image.slidesharecdn.com/surgicalcatalog-121209110056-phpapp02/95/surgical-and-dental-instruments-12-638.jpg?cb=1392649304>

Rongeur

Used to cut and extract tissue and is distinguished by having a spring-loaded hinge. The tips are cupped or beveled and the edges are sharp.

Types of rongeurs

single hinge (single-action rongeur) or two hinges (double-action rongeur).

A double-action rongeur creates twice the leverage of a single-action rongeur.

Stille rongeur- heavier rongeur used in orthopedic and neurosurgical procedures

Kerrison rongeur,- A long-handled delicate rongeur, often is used in spinal surgery.

Pituitary rongeur- long, fine-tipped rongeur used to remove tissue in areas that are difficult to reach, such as the vertebral column and nasal sinus.



Kerrison rongeur image adress:

<https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcSYJtDLrgHPnmnteowjhjo-ltQ0jsPff3Ljww&usqp=CAU>

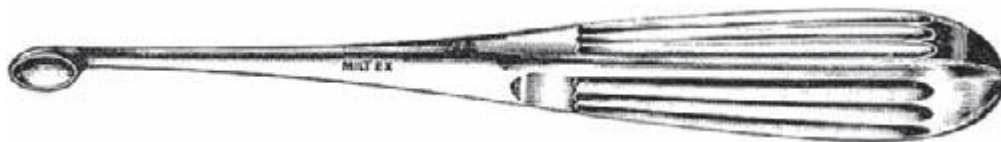
Pictorial link to other types of ronguers: https://orsupply.com/category/876_872_911

Shears

They are large cutting instruments used to sever bone tissue. They are commonly used during thoracic surgery when one or more ribs must be cut for access to the chest cavity.

Curette

It is a small cup with a sharpened, serrated, or smooth rim at the end of the handle used in many specialties for scooping out tissue, including bone and soft tissue. Very fine curettes are used in ear, paranasal, and spinal surgery while larger heavier curettes are used in orthopedic procedures. Soft tissue curettes are used in gynecological surgery for endometrial curettage.



Curette :A courtesy Integra Miltex, York, Pa

Osteotome, Chisel, and Gouge

Chisel - an orthopedic cutting instrument that is used with a mallet (like chisels used in sculpting or carpentry). They are available in many widths and sizes that fit a particular specialty. For example, a chisel used in paranasal surgery is much more delicate than one designed for the tibia. Chisels and osteotomes are often confused with one another yet the osteotome is beveled on both sides, whereas a chisel blade is sloped on one side only. A

large osteotome often is used to remove bone from the iliac crest for use as a graft elsewhere in the body. The chisel produces a straight-sided cut. A gouge is a V-shaped bone chisel whose cut looks like a small trough.



Chisel: A courtesy Integra Miltex, York, Pa

Elevator

It is used to separate or “lift” tissue mostly in separating the periosteum from bone and in vascular surgery. Types of elevators

The heavy, round cutting elevator, such as the Lambotte elevator, slices tissue as it elevates.

The small, square-tipped Key elevator also has a sharp edge but is much more delicate.

Very finely balanced elevators, such as the Penfield and Freer elevators, are used in soft tissue surgery such as neural and vascular procedures.

The joker is a very commonly used elevator. Its short handle and strong tip make it ideal for separating connective tissue layers without causing bleeding

Rasp

It is used to remodel bone. Many sizes, shapes, and surfaces are available.

Fine rasps are used in ear, nose, and throat surgery for the delicate bones and surfaces of the nasal sinuses and ear.

Heavy rasps are used to ream the medullary canal of long bones in preparation for an implant or to prepare the surface of the humerus before sutures are attached.

A rib stripper is a type of rasp used to scrape tough connective tissue from the surface of a rib before it is cut with rib shears. Examples of rib strippers include Matson, Alexander, and Doyen rib rasps.

Saw

They are used in procedures that require bone cutting.

Clamping and Occluding – Are used in many surgical procedures for compressing blood vessels or hollow organs, to prevent their contents from leaking. These instruments are also used to control bleeding. They are either straight, curved or angled, and have a variety of inner jaw patterns.



 alamy stock photo

CFC7YF
www.alamy.com

Image address on theatre clamps: <https://c8.alamy.com/comp/CFC7YF/close-up-of-clamps-surgical-instruments-medical-and-surgical-equipment-CFC7YF.jpg>

Retracting and Exposing – These surgical instruments are used to hold back, or retract organs and tissue so the surgeon has access to the operative area. They spread open the skin, ribs and other tissue; and are also used separate the edges of a surgical incision. Some retracting and exposing instruments are “self-retaining,” meaning they stay open on their own. Others manual styles need to be held open by .

Examples of retractors:

Grasping and Holding – These instruments, as their name suggests, are used to grasp and hold tissue or blood vessels that may be in the way during a surgical procedure. Medical assistants must make sure these instruments are working correctly, and that their locking mechanisms are lubricated for easy opening and closing, particularly after cleaning and reprocessing. Forceps are a very good example of these types of instruments.

The following link provides further details on classification of instruments with examples and pictures: <https://www.surgicalholdings.co.uk/download-catalogue.html>

Operation Theatre Instrument, apparatus and consumables handling techniques

The surgical technologist takes a proactive role in checking instruments for safety and correct operation. This can be done while instruments are being prepared for a procedure, or after the procedure when instruments are sorted and prepared for terminal disinfection. Stainless steel instruments can be damaged mechanically or structurally, or there can be defects on the surface that lead to weakness or cross-infection. Damaged instruments must be withdrawn

from service because they can lead to patient injury, lost operating time, and increased expense. Equipment and instrument management begins in the operating room at the point of instrument use.

General guidelines in handling of instruments include the following:

Inspect all instruments before use on the instruments table and clearly identify any defective or missing instruments.

Handle instruments carefully and one at a time or in small numbers.

Use instruments only for the purpose for which they were intended.

Use scissors only for the material for which they were designed because cutting other materials can cause misalignment, loss of sharpness or damage.

Use forceps appropriate to the task at hand because improper alignment can create serious problems with function and harm to the patient

Match needle holders to the size of needle for which it is intended.

During the procedure, wipe blood and tissue from instruments immediately after use. Debris that is allowed to dry on instruments causes deterioration, corrosion, and pitting. Saline solution may be used to wipe instruments that will be used again during the procedure.

Protect tips of instruments with tip protectors to prevent them contacting other instruments and snagging on towels and sponges

Remove disposable sharps and manage sharps appropriately following the procedure.

During preparation of instruments, open box locks, insure that all multipart instruments are disassembled, keep parts together for easy reassembly after processing, expose all surfaces to come into contact with cleaning, disinfectants, and sterilants

All cannulated equipment and handpieces should be flushed.

Organize instruments into sets- All instruments, used and unused, go back into the set.

Place heavy instruments on the bottom; lighter instruments on top.

Contain contaminated instruments to protect personnel.

Transport instruments to SPD as quickly as possible- Instruments should not remain in water for lengthy periods of time because biofilms may form, particularly within lumens.

Follow the correct procedure for decontamination of instruments as per WHO standards or institutional policy.

Use this link for further details on instruments handling: <https://rocol.com.co/wp-content/uploads/2015/07/CE-ManagingSurgicalInstruments-131209-FINAL.pdf>

Common techniques for inspection of instruments are as outlined in the following video link <https://www.youtube.com/watch?v=CF4WbcqR9RI>

Verification and recording of Operation Theatre Instruments, apparatus and consumables

Different registers are used to verify and maintain a record of operation theatre instruments and other supplies. The unit of competency 'maintain theatre instruments and apparatus' details how to conduct verification of operation theatre instruments, apparatus and consumables.

Disposal of swabs and sharps

In all surgical procedures, all sponges and sharps are counted before, during, and at the end of the procedure. Therefore, before swabs and sharps are disposed, a count must be performed

before swabs and sharps disposal. The following is an outline of the standard procedure for surgical count:-

What

- Soft goods (textiles) including radiopaque sponges of all types, surgical towels, and packing material (e.g., material used in the nasal cavity to absorb blood)
- Individual suture packages
- Sharps including intact knife blades, hypodermic needles, suture needles, trocars, and fragments, if broken
- Instruments
- Miscellaneous items such as electrosurgery tips, cranial (Raney type) clips and their cartridges, umbilical and vessel loops, electrosurgery cleaning pads, small bottles and their caps, medical device parts, and any other object that can be lost in the surgical wound

When

- Before the procedure (to establish a baseline)
- Whenever additional items are introduced to the sterile field intraoperatively
- At the start of wound closure
- Before closing any hollow organ
- Before closing a body cavity
- During closure of skin or other final tissue layer
- Whenever permanent relief personnel join the surgical team
- At the request of the surgeon or any other team member

How

- According to the healthcare institution's policy
- In a systematic, deliberate way, without distraction or interruption
- Without deviation from policy and protocol
- In an established sequence by the type of item being counted (e.g. instruments, sponges, sharps)
- By separating or pointing to each and every item and counting them individually
- Audibly and visually; both people performing the count do so aloud, as they see the items being counted

Who

- As designated by health care facility policy
- The circulator and scrubbed technologist or nurse
- Other members of the sterile team and circulator

All counts are documented in the patient record. Documentation includes names of the individuals who participated in the counts (including relief staff) and their signatures attesting to a correct or incorrect count. Count sheets on which real-time counts are documented may become part of the permanent record, according to facility policy.

A retained item is reported as a sentinel event, even if the item is eventually located and the count validated. This requires documentation on an incident report or by other facility protocol. If the item is found, it is documented as a near miss. The report must include steps taken to find the missing item. Once a count has been complete, swabs and sharps are discarded in relation to OSHA guidelines.

Environmental health

The surgical environment contains many potential sources of environmental hazard. Some of these are obvious, but others are not. Environmental engineering in the operating room follows national medical engineering standards for electrical circuits, inline gases, lighting, and other utilities. Strict safety standards ensure that patients and staff members are protected from extreme hazards and accidents such as fire, explosion, and electrocution.

Learning Activities

Case study

You are participating in a surgical count after skin count. You and your team observe that there is a missing gauze. What will you do as a perioperative theatre technician?

Field Visit to Theatre

Objectives of the visit

To identify instruments and consumables according to the procedure to be carried out

To count instruments, swabs and sharps before and after the procedure according to WHO standards

To confirm and record instruments, swabs, sharps and consumables according to the procedure performed

To discard swabs and sharps according to WHO standards and environmental protection regulation and SOPs

1.2.4.4.3 Self-Assessment

You are provided with the following questions for self -assessment, attempt them and check your responses

Describe ten (10) guidelines for safe handling of theatre instruments and apparatuses.

What is an alexander rasp used for in a surgical operation?

Explain five (5) types of surgical scissors and their specific use in theatre.

What are three shortcomings of using damaged operation theatre instruments and apparatus?

What should be counted in a standard procedure for surgical count?

When should standard surgical counts be performed?

Who should be involved in a surgical count?

What should be documented after a surgical count?

1.2.4.5 Tools, Equipment, Supplies and Materials

Assorted operation theatre instruments: cutting. Holding/grasping, retracting, clamping and occluding instruments

Operation theatre apparatus and consumables such as Branulars, Cannula, Gloves, Scrubs, Blades, swabs

1.2.4.6 References

Linda Timmons & Terri Goodman: Effective management of surgical instrument. Retrieved from: <https://rocol.com.co/wp-content/uploads/2015/07/CE-ManagingSurgicalInstruments-131209-FINAL.pdf> on 7th December 2020.

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Nilsen E: Managing equipment and instruments in the operating room, AORN Journal 81:349, 2005.

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RESPONSES

Describe ten (10) guidelines for handling of operation theatre instruments and apparatuses

Inspect all instruments before use on the instruments table and clearly identify any defective or missing instruments.

Handle instruments carefully and one at a time or in small numbers.

Use instruments only for the purpose for which they were intended.

Use scissors only for the material for which they were designed because cutting other materials can cause misalignment, loss of sharpness or damage.

Use forceps appropriate to the task at hand because improper alignment can create serious problems with function and harm to the patient

Match needle holders to the size of needle for which it is intended.

During the procedure, wipe blood and tissue from instruments immediately after use. Debris that is allowed to dry on instruments causes deterioration, corrosion, and pitting. Saline solution may be used to wipe instruments that will be used again during the procedure.

Protect tips of instruments with tip protectors to prevent them contacting other instruments and snagging on towels and sponges

Remove disposable sharps and manage sharps appropriately following the procedure.

During preparation of instruments, open box locks, insure that all multipart instruments are disassembled, keep parts together for easy reassembly after processing, expose all surfaces to come into contact with cleaning, disinfectants, and sterilants

All cannulated equipment and handpieces should be flushed.

Organize instruments into sets- All instruments, used and unused, go back into the set.

Place heavy instruments on the bottom; lighter instruments on top.

Contain contaminated instruments to protect personnel.

Transport instruments to SPD as quickly as possible- Instruments should not remain in water for lengthy periods of time because biofilms may form, particularly within lumens.

Follow the correct procedure for decontamination of instruments as per WHO standards or institutional policy.

What is an Alexander rasp used for in a surgical operation?

It is used to scrape tough connective tissue from the surface of a rib before it is cut with rib shear.

Explain five (5) types of surgical scissors and their specific use in theatre

Small, sharp-tipped scissors, such as iris scissors, are used for extremely fine dissection in plastic surgery.

Castroviejo scissors are commonly used in microsurgery.

Round-tipped, light dissecting scissors, such as Metzenbaum scissors, are used extensively on delicate tissue in general surgery. They are the most commonly used scissors in general surgery

Fibrous connective tissue requires heavier scissors, such as the curved Mayo scissors.

Straight Mayo scissors are used for cutting suture. NB: tissue scissors must never be used to cut suture because it dulls them quickly.

Wire-cutting scissors are used for stainless steel and other metal suture materials.

What are three shortcomings of using damaged operation theatre instruments and apparatus?
they can lead to patient injury

lost operating time,

increased expense.

What should be counted in a standard procedure for surgical count?

Soft goods (textiles) including radiopaque sponges of all types, surgical towels, and packing material (e.g., material used in the nasal cavity to absorb blood)

Individual suture packages

Sharps including intact knife blades, hypodermic needles, suture needles, trocars, and fragments, if broken

Instruments

Miscellaneous items such as electrosurgery tips, cranial (Raney type) clips and their cartridges, umbilical and vessel loops, electrosurgery cleaning pads, small bottles and their caps, medical device parts, and any other object that can be lost in the surgical wound

When should standard surgical counts be performed?

Before the procedure (to establish a baseline)

Whenever additional items are introduced to the sterile field intraoperatively

At the start of wound closure

Before closing any hollow organ

Before closing a body cavity

During closure of skin or other final tissue layer

Whenever permanent relief personnel join the surgical team

At the request of the surgeon or any other team member

Who should be involved in a surgical count?

The circulator

scrubbed technologist or nurse

Other members of the sterile team

What should be documented after a surgical count?

names of the individuals who participated in the counts

their signatures

Date and time of the count

An attest to a correct or incorrect count.

Learning Outcome 5 Respond to Operation Theatre alarms and signals

Introduction to the learning outcome

This learning outcome specifies the content of competencies required to plan for Execution of Perioperative Theatre Safety Precautions. It includes types of theatre alarms, Identifying theatre alarms and signals, Analysis of alarms and signals, Responding to alarms and signals, Operation Theatre Practice

1.2.5.2 Performance Standard

- I. Theatre alarm signals are identified in accordance with type and source of alarm
- II. Theatre alarms signals are received and analysed according to type and source of alarm
- III. Signals are acted upon according to work place policy and SOPs

1.2.5.5 Information Sheet

Definition of terms

Alarm- a system designed to signal the presence of a hazard requiring urgent attention and to summon emergency medical personnel

Signal- something that conveys information or causes an action.

Types of theatre alarms

Alarm systems are designed to alert staff members to safety risks and should never be turned off or made barely audible. Alarms are key components of peri-operative monitoring devices and function as signals for life-threatening situations as well as device malfunction or imminent danger. An alarm is an automatic warning that results from a measurement, or any other acquisition of descriptors of a state, and indicates a relevant deviation from a normal state.

Common operation theatre alarms

Shouting Alarms

Bell alarm

Electronic alarms

The following link provides further description of common perioperative theatre alarms

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3672485/table/T1/?report=objectonly>

Identifying theatre alarms and signals

The Association of Theatre Technologists developed guidelines to support healthcare delivery organizations reinforce best practices in alarm management as related to the role and duties of the Certified Surgical Technologist (perioperative theatre technologist).

The surgery department should develop a systematic, coordinated approach to the safe management of alarms that involves the surgical team, information technology experts, healthcare technology management technicians (biomedical technicians), risk management and health delivery organization administration.

The surgery department should complete a thorough review of an adverse event involving harm or potential harm to a patient and improper management of an alarm(s).

The perioperative theatre technologist is responsible for the control of the active electrode (also called electrosurgical or electrocautery pencil or Bovie) when not in use to prevent

inadvertent activation to avoid burns to the patient and surgical team, and ignition or puncture of the drapes.

The perioperative theatre technologist should complete continuing education to remain current in their knowledge of alarm management.

1.2.5.4 Learning Activities

1.2.5.4.1 Case study

You are performing decontamination of instruments in theatre when you hear a fire alarm. There is an ongoing surgical operation on the theatre table and the patient has received general anesthesia. What steps will you take?

1.2.5.4.2 Field Visit to Theatre

Objectives of the visit

To identify Theatre alarm signals in accordance with type and source of alarm

To receive and analyse theatre alarms signals according to type and source of alarm

To act upon Signals according to work place policy and SOPs

1.2.5.5 Assignment

Read and make notes on:

Analysis of alarms and signals

Responding to alarms and signals

1.2.5.6 Tools, Equipment, Supplies and Materials

Theatre alarms- ventilatory, hemodynamic and others.

Record books

Theatre equipment

A document of institution policies specific to theatre

1.2.5.7 References

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Learning Outcome 6 Evaluate and wind-up perioperative Theatre safety

Introduction to the learning outcome

This learning outcome specifies the content of competencies required to plan for Execution of Perioperative Theatre Safety Precautions. It includes Occupation health and safety Act, Operation Theatre room layout, Documentation of perioperative safety measures ,Waste disposal and Operation Theatre Practice.

1.2.6.2 Performance Standard

1. Safety measures are followed in accordance with Occupational Safety & health Act standards
2. Operation theatre is well set according to Standard Operation Procedures and workplace policy
3. Operation theatre safety measures are documented according to workplace policy
4. Incident report is written as per workplace policy
5. Waste is disposed with due regard to environment protection regulations

1.2.6.3 Information Sheet

Definition of terms

Safety measures- measures taken to reduce harm or improve safety

Standard operating procedures- a set of step-by-step instructions compiled by an [organization](#) to help workers carry out routine operations

Occupation health and safety Act, 2007

Occupational Safety and Health (OSH) is an area concerned with protecting the safety, health and welfare of people engaged in work or employment. OSH targets four basic aspects:

The protection and promotion of workers health by preventing and controlling occupational diseases and accidents

The development and promotion of healthy and safe work, work environments and work organizations

Enhancement of physical, mental and social well-being of workers; and

Enabling workers to conduct socially and economically productive lives and to contribute positively to sustainable development (WHO 2010).

Responsibility of an organization's management

To ensure that the Occupational Safety and Health Policy, Organizational Responsibilities and the detailed arrangements concerning the health, safety and welfare of all members of staff, clients, contractors, visitors and students are made known and implemented in theatre

To review the written statement of the general policy with respect to safety and health in the workplace;

To carry out suitable and sufficient risk assessment to safety and health of members of staff students and others who within their area of responsibility may be affected by activities in the health sector.

Where hazards are identified which cannot be eliminated, the manager will ensure that safe systems of work are designed and implemented, recorded, monitored and reviewed as necessary and results of the assessments are disseminated to the relevant persons;

Will carry out suitable and sufficient risk assessment to safety and health of members of staff students and others who within their area of responsibility may be affected by activities in the health sector.

Where hazards are identified which cannot be eliminated, the manager will ensure that safe systems of work are designed and implemented, recorded, monitored and reviewed as necessary and results of the assessments are disseminated to the relevant persons;

Will investigate all accidents, incidents or near misses concerning health and safety with the view to identifying the cause(s) and preventing a recurrence;

Will participate in regular Occupations and Safety inspections of their areas of responsibility in accordance with agreed procedures

Employee's responsibility

Ensure their own safety, health, and that of persons who may be affected by their act of commission or omission.

Cooperate with the employer and co-workers in their workplace to achieve safe and healthy work environment by following OSH policy guidelines and standard operating procedures

Make themselves familiar with the OSH policy guidelines

Use PPE as provided by employer to prevent risk to his/her safety and health

Comply with safety and health Standard Operating Procedures (SOPs)

Report to the supervisor of possible hazards

Operation Theatre room layout

The layout of the surgical department is based primarily on the principles of infection control—confining and containing areas and sources of contamination.

The design of any surgery department revolves around environmental control, traffic control, and the desire to prevent surgical site infection (SSI). Such factors as the separation of clean and soiled work areas and areas of the department specified as restricted and unrestricted assist in the promotion of this idea. Efficiency is increased with strategic placement of computers, preparation areas, and staff areas.



Image address for the above operation room: <https://www.midhillhospital.org/wp-content/uploads/2016/11/theatre-surgery-the-device-685x313.jpg>

Different Zones/ Areas

Regardless of the design chosen by the facility, certain principles apply universally. Traffic control follows predetermined traffic patterns that all persons entering the department are expected to follow. The OR department is divided into four zones based on varying degrees of cleanliness, in which the bacteriological count progressively diminishes from the outer to the inner zones (operating area) and is maintained by a differential decreasing positive pressure ventilation gradient from the inner zone to the outer zone.

- 1.Unrestricted zone or clean area.
- 2.Semi restricted zone or sub sterile area
- 3.Restricted zone or sterile area
- 4.Disposal zone or area

Unrestricted/ Clean zone or clean area- This is the area where hospital personnel, OT personnel, patient's & their attendants can move about in street clothes. Connects protective zone to aseptic zone and has other areas also like Stores & cleaner room, equipment store room, maintenance workshop, kitchenette, firefighting device room, emergency exit.

Semi restricted/ Protective zone or sub-sterile area- After changing from street clothes, with clean gown, cap & OT slipper, the OT team or personnel are, enter in this area. (Operating room attire is required). It includes - Change rooms for all medical and paramedical staff with conveniences, Transfer bay for patient, material & equipment's, Rooms for administrative staff, Stores & records, Pre & post-operative rooms.I.C.U. and P.A.C.U.

Restricted/Aseptic zone or sterile area: This zone has operation theatres & operation room where operations are done. This is a place where staff & patients in street clothes & shoes are not allowed to enter. Scrubbed personnel wear sterile gowns & gloves & get ready for operation procedure. (Operating room attire is required)

Disposal zone or area- Dirty utility area, Disposal corridor, OT attire mandatory in this area
Sub areas:-

Nurses station or counter - Nurses station is an area where nurses and other health care staff sit behind when not working directly with patients and can perform some of their duties. The station has a counter that can be approached by visitors and patients who wish to receive attention from the nurses. It should be spacious enough to allow two or more personnel to work together.

Pre-operative check area (reception) - It is the area where OT nursing personnel receive patients. Here patients and his records are checked.

Holding area - This area is planned for IV line insertion, preparation, catheter / gastric tube insertion, connection of monitors, & shall have O2 and suction lines. Facility for CPR should be available in this area.

Induction/Anaesthetic room - It should have all facilities as in OT, but there is controversy as to its need. It should provide space for anaesthetic trolleys and equipment and should be located with direct access to circulation corridors and ready access to the operating room. It will also allow cleaning, testing and storing of anaesthesia equipment. It should have sufficient power outlets and medical gas panels for testing of equipment.

Post anaesthetic care units (PACU)/Post-operative or recovery room – Room where patients are kept & cared until they are out of anaesthesia & until their vital signs become stable. These should contain a medication station, hand washing station, nurse station, storage space for stretchers, supplies and monitors / equipment and gas, suction outlets and ventilator.

Staff room – Men and women change dress from street cloth to OT attire; lockers and lavatory are essential; rest room etc. are desirable.

Sanitary facility for staff- One washbasin and one western closet (WC) should be provided for 8-10 persons. Showers and their number is a matter of local decision. Inclusion of toilet facilities in changing rooms is not acceptable; they should be located in an adjacent space.

Offices – for staff nurse and anaesthesia staff- The office should allow access to both unrestricted and semi-restricted areas as frequent communication with public is needed.

Rest rooms- Pleasant and quiet rest for staff should be arranged either as one large room for all grades of staff or as separate rooms; both have merits. Comfortable chairs, one writing table, a book case etc., may be arranged.

Laboratory – Small laboratory with refrigerator for pathologist to be arranged. □

Store room

Scrub room- This is planned to be built within the restricted area. This should be spacious. Scrub area should have deep & wide sink to avoid splash of water on the surrounding area. There should be facilities for running water. The taps should be such that they could be opened or closed by foot pedals or they have long handles that can be operate by elbows.

Waiting room – Patient's attendants or family member's waiting room with attached toilet & drinking water facility. □

Utility Room - This is also called clean up room where instruments, articles, gloves are received following surgical procedures. Here all these are cleaned, washed & wiped dry & then wrapped, stored & sent for sterilization.

Disposal or sluice room - This room receives all soiled linen & dressing. Here all disposable items are separated according to biochemical waste management code & then packed in leak proof containers. The soiled linens are sent to the laundry. The OT set up should be such that it should be possible to dispose off soiled items without taking them through sub-sterile or sterile areas.

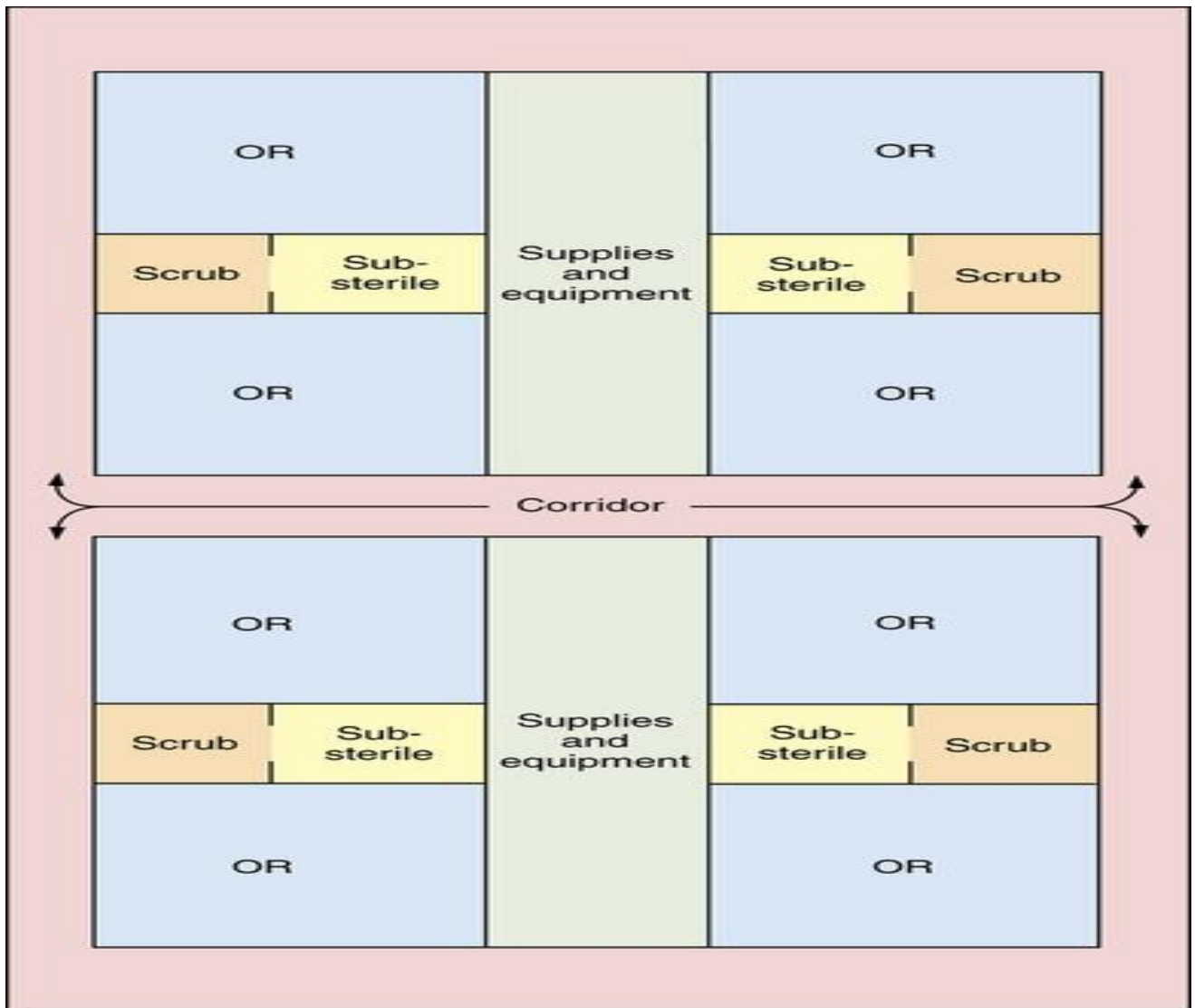


Image address to the above operation theatre zones: https://basicmedicalkey.com/wp-content/uploads/2017/04/B978032307358500019X_f10-04-9780323073585.jpg

Documentation of perioperative safety measures

Documentation for the surgical procedure is completed as usual, with accurate recording of the chain of events. Operative records for patient care, anesthesia, sponge counts, and all usual forms must be completed as for any case.

Assignment: Review guidelines on documentation of perioperative safety measures

Perioperative theatre practice

Recommended practices are optimum behavioral objectives for caregivers. They may not always be achievable, as standards are, because of limitations in a particular practice setting. Recommended practices state what ideally can be done. Practice are directed toward providing safety in the perioperative environment. They are premised in principles of microbiology, scientific literature, validated research, evidence-based practice, and experts' opinions. Although compliance is voluntary, individual commitment, professional conscience, and the practice setting should guide perioperative caregivers in using these recommended practices. They represent an optimal level of practice and are achievable

Sharps, swabs and instruments

It is recommended that counts of swabs, instruments (including scratchpads and diathermy tips), and sharps (needles and blades) should be undertaken at strategic points during an operation. This is to ensure no foreign body remains in a patient after closure of a wound.

The scrub practitioner is responsible for counting all items used in an operation before incision, at first layer closure, and at skin closure. It is also standard practice to give a verbal report to the surgeon at the end of the procedure to the effect that all sharps, swabs, and instruments are correct. The count requires full concentration, should be audible, and should be carried out by two people, one of whom must be qualified. Both people should sign the theatre register.

The initial count should be recorded before the operation begins and any items that are added during the course of the operation should be recorded. All items counted during the course of the operation should be recorded clearly on a dry-wipe board. Items removed from the sterile field should be visibly deleted with a scoring line or cross. Each swab counted must be checked to ensure its radiopaque strip is intact. This is because, in the event of a lost swab (or sharp, or instrument), the surgeon might request an X-ray to ensure that it has not been left in the patient. Any item deliberately left inside the patient should be clearly documented in the patient's notes.

Every sharp object in a theatre should be regarded as a potential source of injury. Accepted practices in the handling of sharps should be adhered to, for example avoiding manual re-sheathing of needles, and disposing of sharps in containers. Needles, blades or instruments should never be passed directly hand to hand, but should be passed via a 'neutral zone', for example a receiver. Needles and blades should be mounted for use using the correct instrument, never the fingers. Any sharps injury should be managed and reported to occupational health according to local policy.

Waste disposal

Parker (1999) explains that theatre design should incorporate a sequence of clean zones from the entrance to the operating area, to allow easy movement of staff from one clean area to another. Conversely, waste and contaminated items should be removed through 'dirty' areas without passing through clean areas. Body fluids collected in suction bottles should be treated with a gelling agent and allowed to solidify before being double-wrapped to prevent leakage (see case study, Box 2), as should waste tissue and bone fragments. All clinical waste (placed in yellow bags) should be labelled with the date, theatre number and case number to allow traceability.

From an environmental point of view, machines are now available that use microwaves to make used materials safe for disposal in landfill sites. Normal rubbish should be placed in black bags and disposed of in landfill sites to avoid incineration and minimise air pollution.

1.2.6.4 Learning Activities

1.2.6.4.1 Case study

A junior staff reports to you, theatre manager, with complaints of having sustained a deep cut while clearing cutting instruments following an orthopedic surgery. What will be your response?

1.2.6.4.2 Field Visit to Theatre

Objectives of the visit

To follow Safety measures in accordance with Occupational Safety & health Act standards

To set Operation theatre according to Standard Operation Procedures and workplace policy

To document Operation theatre safety measures according to workplace policy

To write Incident report as per workplace policy

To dispose waste with due regard to environment protection regulations

Self-Assessment

You are provided with the following questions for self -assessment, attempt them and check your responses

What four aspects does Occupational Safety and Health (OSH) target?

Describe the responsibility of an organization's management in OSH?

What is the specific responsibility of an employee in OSH

What are three (3) factors guide the layout of an operation theatre?

What activities are the sub areas of Unrestricted/ Clean zone or clean area?

Why is it important to take surgical counts before skin closure in an operation?

What documentation should be made on the clinical waste to ensure theatre safety?

What is the difference between decontamination and sterilization?

Tools, Equipment, Supplies and Materials

Waste disposal bins

Sharps boxes

Decontamination solutions

Decontamination boxes

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RESPONSES

What four aspects does Occupational Safety and Health (OSH) target?

The protection and promotion of workers health by preventing and controlling occupational diseases and accidents

The development and promotion of healthy and safe work, work environments and work organizations

Enhancement of physical, mental and social well-being of workers; and

Enabling workers to conduct socially and economically productive lives and to contribute positively to sustainable development (WHO 2010).

Describe the responsibility of an organization's management in OSH

To ensure that the Occupational Safety and Health Policy, Organizational Responsibilities and the detailed arrangements concerning the health, safety and welfare of all members of staff, clients, contractors, visitors and students are made known and implemented in theatre

To review the written statement of the general policy with respect to safety and health in the workplace;

To carry out suitable and sufficient risk assessment to safety and health of members of staff students and others who within their area of responsibility may be affected by activities in the health sector.

Where hazards are identified which cannot be eliminated, the manager will ensure that safe systems of work are designed and implemented, recorded, monitored and reviewed as necessary and results of the assessments are disseminated to the relevant persons;

Will carry out suitable and sufficient risk assessment to safety and health of members of staff students and others who within their area of responsibility may be affected by activities in the health sector.

Where hazards are identified which cannot be eliminated, the manager will ensure that safe systems of work are designed and implemented, recorded, monitored and reviewed as necessary and results of the assessments are disseminated to the relevant persons;

Will investigate all accidents, incidents or near misses concerning health and safety with the view to identifying the cause(s) and preventing a recurrence;

Will participate in regular Occupations and Safety inspections of their areas of responsibility in accordance with agreed procedures

What is the specific responsibility of an employee in OSH

Ensure their own safety, health, and that of persons who may be affected by their act of commission or omission.

Cooperate with the employer and co-workers in their workplace to achieve a safe and healthy work environment by following OSH policy guidelines and standard operating procedures

Make themselves familiar with the OSH policy guidelines

Use PPE as provided by employer to prevent risk to his/her safety and health

Comply with safety and health Standard Operating Procedures (SOPs)

Report to the supervisor of possible hazards

What three (3) factors guide the layout of an operation theatre?

environmental control

traffic control

desire to prevent surgical site infection (SSI).

What activities are the sub areas of Unrestricted/ Clean zone or clean area?

Stores

equipment store room

maintenance workshop

Kitchenette

firefighting device room

Why is it important to take surgical counts before skin closure in an operation?

This is to ensure no foreign body remains in a patient after closure of a wound..

What documentation should be made on the clinical waste to ensure theatre safety?

It should be labelled with the date, theatre number and case number to allow traceability.

What is the difference between decontamination and sterilization?

Decontamination is the process of elimination of pathogens except the spore from inanimate things while sterilization is the process that eliminates all micro-organisms such as bacteria, viruses, fungi and parasites, including bacterial endospores.